

DANIELE PESSINA

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EDUCATION

Imperial College London

2022 – Present

PhD in Chemical Engineering - CASE Studentship with AstraZeneca

- Thesis title: A hybrid *in-silico/in-vitro* approach for integrated design and optimisation of large-molecule crystallisation

Imperial College London

2018 – 2022

MEng in Chemical Engineering, 1st Class Honours

EXPERIENCE

PhD Researcher, Process Systems Engineering

2022 - Present

Supervision: Dr. Maria Papathanasiou & Prof. Jerry Y. Y. Heng

- Developed and deployed novel ML architectures (neural ODEs, transformers, autoencoders), transfer learning, reinforcement learning and uncertainty quantification methods in JAX for improved time-series modelling of sparse and multi-fidelity datasets.
- Built autodifferentiable probabilistic inference pipelines in JAX and Julia for ODE and PDE -based numerical simulations, which enabled uncertainty-aware parameter estimation, robust system optimisation and flexibility assessments.
- Designed feature selection and model reduction workflows via sensitivity analysis that enhanced identifiability and tractability in crystallisation process models.
- Investigated explainability methods (symbolic regression, SHAP, sensitivity analysis), improving interpretability of ML surrogates of complex and intractable DMPK pharmacokinetic models.
- Conducted *in-vitro* medium-scale lysozyme crystallisation experiments with MT EasyMax and PAT tools, generating experimental evidence for template-induced protein nucleation mechanisms.

Consultant Data Scientist

March - July 2025

Haleon

- Delivered ML-based soft sensors providing real-time product quality estimation from high-frequency manufacturing data, reducing reliance on offline batch testing.
- Engineered time-aggregated features and evaluated sliding/increasing window strategies for online viscosity predictions, supporting proactive adjustments in downstream filling and packaging.
- Benchmarked LSTM and transformer models against classical feature-engineered approaches, improving adaptability to new product families and strengthening robustness under imbalanced datasets.

Lixea - Summer intern

June - Sep. 2021

Supervision: Prof. Jason Hallett

- Developed a techno-economic simulation model for large-scale biomass fractionation, quantifying process uncertainty and capital cost variability, and provided strategic forecasts directly used in investor reports.

TECHNICAL SKILLS

Core: SciML, probabilistic programming, ODE/PDE numerical modelling, Bayesian inference and optimisation, uncertainty quantification, time-series modelling, surrogate modelling, optimization under uncertainty, RL, distributed computing, GPU acceleration

Software: Julia, JAX, Python, Pytorch, Git, Databricks, MLflow

Lab: Crystallisation equipment (Mettler-Toledo EasyMax, FTIR, UV-Vis, PAT tools), mammalian cell culture

Languages: Italian (native), English (fluent), Spanish (conversational)

PUBLICATIONS

Pessina, D. *et al.* (2025) Transfer Learning Assessment of Data-driven Crystallisation Processes via Constrained Neural Ordinary Differential Equations; *bioRxiv*

Pessina, D. *et al.* (2025) Model-based approach to template-induced macromolecule crystallisation; *Systems & Control Transactions*

Pessina, D. *et al.* (2025) Machine learning-enhanced Sensitivity Analysis for Complex Pharmaceutical Systems; *Systems & Control Transactions*

Pessina, D. *et al.* (2025) Integrated In Vitro/In Silico Uncertainty Quantification Method for Protein Crystallization Models; *Industrial & Engineering Chemistry Research*